

Lecture #7

PROPOSITIONAL LOGIC

Boolean Functions, Equivalences, Complete Sets of Functions,
Normal Forms (CNF, DNF)

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Boolean functions

Definition. Any function $f: \{0,1\}^n \rightarrow \{0,1\}$ (with inputs and output are only 0 or 1) is called ***Boolean function***.

Logical functions can be represented as Boolean functions, if

- 1 is mean true statement, and
- 0 is mean false statement

Question: How many Boolean functions with n variable can have?

Answer: 2^{2^n}

Boolean functions with 1 variable

The following is all Boolean function with one variable:

- $f_1(0) = 0, f_1(1) = 0;$
- $f_2(0) = 0, f_2(1) = 1;$
- $f_3(0) = 1, f_3(1) = 0;$
- $f_4(0) = 1, f_4(1) = 1.$

These functions are conveniently written as the following table (We will call it a ***Truth table***)

x	f_1	f_2	f_3	f_4
0	$f_1(0) = 0$	$f_2(0) = 0$	$f_3(0) = 1$	$f_4(0) = 1$
1	$f_1(1) = 0$	$f_2(1) = 1$	$f_3(1) = 0$	$f_4(1) = 1$

Boolean functions with 1 variable

The following is all Boolean function with one variable:

- f_1 corresponds to the false function, lets denote it as 0;
- f_2 corresponds to the identity ($f(x) = x$), lets denote it as x ;
- $f_3(0)$ corresponds to the negation of x , lets denote it as $\bar{x}$;
- $f_4(0)$ corresponds to the true function, lest denote it as 1.

x	0	x	$\bar{x}$	1
0	0	0	1	1
1	0	1	0	1

Boolean functions with 2 variables

Truth table of a Boolean function ($f(x, y)$) with 2 variables will consist 4 rows

- first row for value when $x = 0$ and $y = 0$;
- second row for value when $x = 0$ and $y = 1$;
- third row for value when $x = 1$ and $y = 0$;
- fourth row for value when $x = 1$ and $y = 1$.

[illegible]

Boolean functions with 2 variables

[illegible]

Boolean functions with 2 variables

x	y	0			x		y					$\bar{y}$		$\bar{x}$			1
0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1

Boolean functions with 2 variables

A new names for Boolean functions with 2 variables:

1. $x \cdot y$ is conjunction or multiplication of x and y ;
2. $x \vee y$ is disjunction of x and y ;
3. $x + y$ is sum by modulo 2;
4. $x \rightarrow y$ is implication (x implies y);
5. $x \sim y$ is equivalence of x and y ;
6. $x \mid y$ – Sheffer strough (anticonjunction of x and y);
7. $x \downarrow y$ – Peirce arrow (antidusjunction of x and y).

Boolean functions with 2 variables

x	y	0	$x \cdot y$	$\overline{x \rightarrow y}$	x	$\overline{y \rightarrow x}$	y	$x \oplus y$	$x \vee y$
0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1

x	y	$x \downarrow y$	$x \sim y$	$\overline{y}$	$y \rightarrow x$	$\overline{x}$	$x \rightarrow y$	$x y$	1
0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1

Vector form of a Boolean function

Definition. Let $f: \{0,1\}^n \rightarrow \{0,1\}$ be a Boolean function. Fix an ordering of inputs $\alpha_1, \alpha_2, \dots, \alpha_{2^n} \in \{0,1\}^n$ (e.g., lexicographic order). The vector form of f is the binary vector

$$\mathbf{v}(f) = (f(\alpha_1) f(\alpha_2) \dots f(\alpha_{2^n})) \in \{0,1\}^{2^n}$$

Example. For $n = 2$ (order 00, 01, 10, 11), if

$$f(x, y) = x + y$$

then

$$\mathbf{v}(f) = (0110)$$

Example. If vector of the functions f and g are $\mathbf{v}(f) = (0001)$ and $\mathbf{v}(g) = (1101)$, then find the vector of function

$$h(x, y, z) = f(y, g(x, z)) \vee g(y, z)$$

Solution:

a	b	g	f
0	0	1	0
0	1	1	0
1	0	0	0
1	1	1	1

This equals:

$$g(a, b) = a \rightarrow b = \bar{a} \vee b \text{ and } f(a, b) = a \cdot b$$

Then

$$f(y, g(x, z)) = y \cdot (\bar{x} \vee z)$$

Hence,

$$h(x, y, z) = (y \cdot (\bar{x} \vee z)) \vee (\bar{y} \vee z)$$

Example. If vector of the functions f and g are $\mathbf{v}(f) = (0001)$ and $\mathbf{v}(g) = (1101)$, then find the vector of function

$$h(x, y, z) = f(y, g(x, z)) \vee g(y, z)$$

Solution: $h(x, y, z) = (y \cdot (\bar{x} \vee z)) \vee (\bar{y} \vee z).$

x	y	z	$\bar{x} \vee z$	$y \cdot (\bar{x} \vee z)$	$\bar{y} \vee z$	$h(x, y, z)$
0	0	0	1	0	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	1
1	0	0	0	0	1	1
1	0	1	1	0	1	1
1	1	0	0	0	0	0
1	1	1	1	1	1	1

Answer:

$$\mathbf{v}(h) = (11111101)$$

Example. Build Truth Table for the function
 $((x \mid y) \mid z) \rightarrow x$

Solution:

x	y	z	$x \mid y$	$(x \mid y) \mid z$	$((x \mid y) \mid z) \rightarrow x$
0	0	0	1	1	0
0	0	1	1	0	1
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	1	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	1	0	1	1

The main equivalences

Definition. Two Boolean functions f and g is called equivalence (or equal, and we denote $f = g$) if they have same values for any inputs (or same Truth table).

The Main Equivalences:

1. All the following operations are commutative $\{\cdot, \vee, +, \sim, |, \downarrow\}$
2. All the following operations are associative $\{\cdot, \vee, +, \sim\}$
3. Distributive laws:
 - $(x \vee y) \cdot z = (x \cdot z) \vee (y \cdot z);$
 - $(x \cdot y) \vee z = (x \vee z) \cdot (y \vee z);$
 - $(x + y) \cdot z = (x \cdot z) + (y \cdot z).$

The main equivalences

4. De Morgan laws:

$$\blacksquare \overline{x \cdot y} = \bar{x} \vee \bar{y}$$

$$\blacksquare \overline{x \vee y} = \bar{x} \cdot \bar{y}$$

5. Absorption laws:

$$\blacksquare x \vee (x \cdot y) = x;$$

$$\blacksquare x \cdot (x \vee y) = x.$$

$$6. x \rightarrow y = \bar{x} \vee y$$

$$7. x \sim y = \overline{x + y} = (x \cdot y) \vee (\bar{x} \cdot \bar{y}) = (x \vee \bar{y}) \cdot (\bar{x} \vee y)$$

$$8. x \mid y = \overline{x \cdot y} = \bar{x} \vee \bar{y}$$

$$9. x \downarrow y = \overline{x \vee y} = \bar{x} \cdot \bar{y}$$

Example. Show that the following functions are equivalent

$$f(x, y, z) = (\overline{z \mid y}) \mid ((z \downarrow x) \vee \bar{z})$$
$$g(x, y, z) = (x \downarrow (y \rightarrow x)) \rightarrow (y \vee z)$$

Solution: Simplify $f(x, y, z)$

■ $(\overline{z \mid y}) = \overline{\bar{z} \cdot \bar{y}} = z \cdot y$

■ $z \downarrow x = \overline{z \vee x} = \text{De - Morgan} = \bar{z} \cdot \bar{x}$

■ $(z \downarrow x) \vee \bar{z} = (\bar{z} \cdot \bar{x}) \vee \bar{z} = \text{absorption} = \bar{z}$

$$f(x, y, z) = (\overline{z \mid y}) \mid ((z \downarrow x) \vee \bar{z}) = (z \cdot y) \mid \bar{z} = \overline{(z \cdot y) \cdot \bar{z}} = \bar{0} = 1$$

Simplify $g(x, y, z)$

■ $y \rightarrow x = \bar{y} \vee x$

■ $x \downarrow (y \rightarrow x) = x \downarrow (\bar{y} \vee x) = \overline{x \vee (\bar{y} \vee x)} = \overline{x \vee \bar{y}} = \bar{x} \cdot y$

$$g(x, y, z) = (x \downarrow (y \rightarrow x)) \rightarrow (y \vee z) = (\bar{x} \cdot y) \rightarrow (y \vee z) = \overline{\bar{x} \cdot y} \vee (y \vee z)$$
$$= x \vee \bar{y} \vee y \vee z = x \vee 1 \vee z = 1$$

Functionally Complete Sets of Boolean Functions

Definition. A set $\mathcal{F}$ of Boolean functions is called *functionally complete* (or complete) if every Boolean function can be obtained from functions in $\mathcal{F}$ by superposition.

Example.

1. $\{\neg, \cdot\}$ is complete.
2. $\{\neg, \vee\}$ is complete.
3. $\{\downarrow\}$ (NAND) is complete.
4. $\{\uparrow\}$ (NOR) is complete.
5. $\{\cdot, +\}$

DNF and CNF of Boolean functions

Definition. If x is a variable, then both x and $\bar{x}$ we call *literal*.

A Boolean formula is in ***DNF (Disjunctive Normal Form)***, if it has form

$$K_1 \vee K_2 \vee \cdots \vee K_m$$

where each K_i is a conjunction of literals.

A Boolean formula is in ***CNF (Conjunctive Normal Form)***, if it has form

$$K_1 \cdot K_2 \cdot \cdots \cdot K_m$$

where each K_i is a disjunction of literals.

Perfect (Full) DNF and CNF

Definition.

Perfect Disjunctive Normal Form (PDNF).

A DNF is called *perfect* (or full) if

- each conjunction (minterm) contains all variables,
- each variable appears exactly once (either x_i or $\bar{x}_i$).

Perfect Conjunctive Normal Form (PCNF).

A CNF is called *perfect* if

- each disjunction (maxterm) contains all variables,
- each variable appears exactly once.

Perfect (Full) DNF and CNF

Theorem. Every Boolean function has a unique PDNF and PCNF (up to reordering of terms).

Example. Find DNF, CNF, PDNF and PCNF of the following function

$$(x \vee y) \mid \overline{(z \sim \bar{x})}$$

Solution: $(x \vee y) \mid \overline{(z \sim \bar{x})}$

x	y	z	$x \vee y$	$\bar{x}$	$z \sim \bar{x}$	$\overline{(z \sim \bar{x})}$	$(x \vee y) \mid \overline{(z \sim \bar{x})}$
0	0	0	0	1	0	1	1
0	0	1	0	1	1	0	1
0	1	0	1	1	0	1	0
0	1	1	1	1	1	0	1
1	0	0	1	0	1	0	1
1	0	1	1	0	0	1	0
1	1	0	1	0	1	0	1
1	1	1	1	0	0	1	0

Solution: $(x \vee y) \mid \overline{(z \sim \bar{x})}$

x	y	z	$x \vee y$	$\bar{x}$	$z \sim \bar{x}$	$\overline{(z \sim \bar{x})}$	$(x \vee y) \mid \overline{(z \sim \bar{x})}$
0	0	0	0	1	0	1	1
0	0	1	0	1	1	0	1
0	1	0	1	1	0	1	0
0	1	1	1	1	1	0	1
1	0	0	1	0	1	0	1
1	0	1	1	0	0	1	0
1	1	0	1	0	1	0	1
1	1	1	1	0	0	1	0

PDNF: Take rows where $F = 1$: 000, 001, 011, 100, 110

$$(\bar{x} \cdot \bar{y} \cdot \bar{z}) \vee (\bar{x} \cdot \bar{y} \cdot z) \vee (\bar{x} \cdot y \cdot z) \vee (x \cdot \bar{y} \cdot \bar{z}) \vee (x \cdot y \cdot \bar{z})$$

Solution: $(x \vee y) \mid \overline{(z \sim \bar{x})}$

x	y	z	$x \vee y$	$\bar{x}$	$z \sim \bar{x}$	$\overline{(z \sim \bar{x})}$	$(x \vee y) \mid \overline{(z \sim \bar{x})}$
0	0	0	0	1	0	1	1
0	0	1	0	1	1	0	1
0	1	0	1	1	0	1	0
0	1	1	1	1	1	0	1
1	0	0	1	0	1	0	1
1	0	1	1	0	0	1	0
1	1	0	1	0	1	0	1
1	1	1	1	0	0	1	0

PCNF: Take rows where $F = 0$: 010, 101, 111

$$(x \vee \bar{y} \vee z) \cdot (\bar{x} \vee y \vee \bar{z}) \cdot (\bar{x} \vee \bar{y} \vee \bar{z})$$

Solution: $(x \vee y) \mid \overline{(z \sim \bar{x})}$

DNF: Simplify PDNF

Group terms

$$\begin{aligned} & (\bar{x} \cdot \bar{y} \cdot \bar{z}) \vee (\bar{x} \cdot \bar{y} \cdot z) \vee (\bar{x} \cdot y \cdot z) \vee (x \cdot \bar{y} \cdot \bar{z}) \vee (x \cdot y \cdot \bar{z}) \\ &= [\bar{x} \cdot \bar{y} \cdot (\bar{z} \vee z)] \vee (\bar{x} \cdot y \cdot z) \vee [x \cdot \bar{z}(\bar{y} \vee y)] \\ &= (\bar{x} \cdot \bar{y} \cdot 1) \vee (\bar{x} \cdot y \cdot z) \vee (x \cdot \bar{z} \cdot 1) = \bar{x}\bar{y} \vee \bar{x}yz \vee x\bar{z} \end{aligned}$$

CNF: Simplify PCNF

Group terms

$$\begin{aligned} & (x \vee \bar{y} \vee z) \cdot (\bar{x} \vee y \vee \bar{z}) \cdot (\bar{x} \vee \bar{y} \vee \bar{z}) = (x \vee \bar{y} \vee z) \cdot [\bar{x} \vee \bar{y} \vee (\bar{z} \cdot z)] \\ &= (x \vee \bar{y} \vee z) \cdot [\bar{x} \vee \bar{y} \vee 0] = (x \vee \bar{y} \vee z) \cdot (\bar{x} \vee \bar{y}) \end{aligned}$$

Precedence of Boolean Operations

When parentheses are omitted, Boolean operations are evaluated according to the following priority:

$$1 \bar{x}$$

$$2 x \cdot y$$

$$3 x \vee y$$

$$4 x + y$$

$$5 x \rightarrow y, x \sim y$$

Example.

$$\begin{aligned}\bar{x} \vee y \cdot z &= \bar{x} \vee (y \cdot z) \\ x \rightarrow y \vee z &= x \rightarrow (y \vee z)\end{aligned}$$

Zhegalkin Polynomial (Algebraic Normal Form)

Theorem. Every Boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$ can be uniquely represented in the form

$$f(x_1, \dots, x_n) = a_0 \oplus \sum_i a_i x_i \oplus \sum_{i < j} a_{ij} x_i x_j \oplus \dots \oplus a_{12\dots n} x_1 x_2 \dots x_n$$

where $a_0, a_i, a_{ij}, \dots \in \{0,1\}$

This representation is called the ***Zhegalkin polynomial*** (or Algebraic Normal Form, ANF).

Computing Zhegalkin coefficients (Triangle Method)

Idea. Write the vector of values of f in lexicographic order. Construct a triangular table by taking pairwise sums modulo 2.

Example. Let $n = 2$, order 00, 01, 10, 11, and $\mathbf{v}(f) = (0, 1, 1, 0)$.

$$\begin{array}{cccc} 0 & 1 & 1 & 0 \\ & 1 & 0 & 1 \\ & & 1 & 1 \\ & & & 0 \end{array} \quad (\text{all sums mod } 2)$$

Rule. The leftmost elements of each row are the coefficients of the Zhegalkin polynomial (ANF): a_0, a_1, a_2, a_{12}

Result. Here we obtain $f(x_1, x_2) = x_1 \oplus x_2$.

Example. Find Zhegalkin polynomial of the following function

$$x \rightarrow ((y \vee z) \downarrow x)$$

Solution: Simplify

$$\begin{aligned} x \rightarrow ((y \vee z) \downarrow x) &= \bar{x} \vee ((y \vee z) \downarrow x) = \bar{x} \vee (\overline{(y \vee z)} \cdot \bar{x}) \\ &= \bar{x} \vee (\bar{y} \cdot \bar{z} \cdot \bar{x}) = \text{absorption law} = \bar{x} \end{aligned}$$

x	y	z	$\bar{x}$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

1 1 1 1 0 0 0 0
 0 0 0 1 0 0 0 0
 0 0 1 1 0 0
 0 1 0 1 0
 1 1 1 1
 0 0 0
 0 0
 0

The Zhegalkin coefficients are
 1 0 0 0 1 0 0 0

Monomials in order:

1, z, y, yz, x, xz, xy, xyz

Now attach coefficients:

$$F = 1 \oplus x$$